Modeling and Dynamical Analysis Project

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1 Preparation

To linearize the nonlinear state-space model about the stationary mode $X_0 = [v_0, 0, 0, 0]^T$, we perform a first-order Taylor expansion. The system is given by:

$$\dot{X} = f(X, u)$$

where $X = [v, w, q, \theta]^T$ and $u = \delta_e$. The linearized model takes the form

$$\dot{X} = AX + Bu$$

where:

$$A = \frac{\partial f}{\partial X}(X_0, u_0), \quad B = \frac{\partial B}{\partial X}(X_0, u_0)$$

From the given nonlinear system:

1. $\dot{v} = -0.0156v + 0.0101w - g\sin(\theta) - qw$
2. $\dot{w} = -0.1367v - 1.9635w + g\cos(\theta) + qv + 0.1351\delta_e$
3. $\dot{q} = 0.000948v - 0.02555w - 3.6668q - 0.14456\delta_e$
4. $\dot{\theta} = q$

We compute the Jacobians at X_0, where $v_0 = 186\,m/s$ and $g = 9.81m/s^2$ is the gravitational acceleration.

$$A = \begin{bmatrix} \frac{\partial \dot{v}}{\partial v} & \frac{\partial \dot{v}}{\partial w} & \frac{\partial \dot{v}}{\partial q} & \frac{\partial \dot{v}}{\partial \theta} \\ \frac{\partial \dot{w}}{\partial v} & \frac{\partial \dot{w}}{\partial w} & \frac{\partial \dot{w}}{\partial q} & \frac{\partial \dot{w}}{\partial \theta} \\ \frac{\partial \dot{q}}{\partial v} & \frac{\partial \dot{q}}{\partial w} & \frac{\partial \dot{q}}{\partial q} & \frac{\partial \dot{q}}{\partial \theta} \\ \frac{\partial \dot{\theta}}{\partial v} & \frac{\partial \dot{\theta}}{\partial w} & \frac{\partial \dot{\theta}}{\partial q} & \frac{\partial \dot{\theta}}{\partial \theta} \end{bmatrix} = \begin{bmatrix} -0.0156 & 0.0101 & -w_0 & -g\cos(\theta_0) \\ -0.1367 & -1.9635 & v_0 & -g\sin(\theta_0) \\ 0.000948 & -0.02555 & -3.6668 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Since we are linearizing about $X_0 = [v_0, 0, 0, 0]^T$, we set $w_o = 0$ and $\theta_0 = 0$ and obtain:

$$A = \begin{bmatrix} -0.0156 & 0.0101 & 0 & -g \\ -0.1367 & -1.9635 & v_0 & 0 \\ 0.000948 & -0.02555 & -3.6668 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Let's now compute B:

$$B = \begin{bmatrix} \dfrac{\partial \dot{v}}{\partial \delta_e} \\ \dfrac{\partial \dot{v}}{\partial \delta_e} \\ \dfrac{\partial \dot{v}}{\partial \delta_e} \\ \dfrac{\partial \dot{v}}{\partial \delta_e} \end{bmatrix} = \begin{bmatrix} 0 \\ 0.1351 \\ -0.14456 \\ 0 \end{bmatrix}$$

Since the output equation is $y = X$, we have:

$$C = I_{4\times 4}, \qquad D = 0_{4\times 1}$$

Let's now check the controllability of the system computing the rank of the controllability matrix. We set up the problem writing matrices A, B, C, D on MatLab and then computing the rank of the controllability matrix using the command: `rank(ctrb(Co))` as shown in the snippet of Fig.1.

```
% Compute the controllability matrix
Co = ctrb(A, B);

% Compute the rank of the controllability matrix
rank_Co = rank(Co);

% Display results
disp('Rank of the controllability matrix:');
disp(rank_Co);

% Check if the system is controllable
if rank_Co == size(A,1)
    disp('The system is controllable.');
else
    disp('The system is not fully controllable.');
end
```

Figure 1: Computing Controllability

- If `rank_Co == 4`, the system is fully controllable, meaning the control input δ_e can influence all states.
- If `rank_Co < 4`, the system is not fully controllable, meaning some states cannot be controlled through, and additional control inputs might be needed.

In this case the system is controllable.

To analyze the aircraft's natural modes, we need to compute the eigenvalues of the state matrix A, which represent the system's modes of motion. These eigenvalues determine the system's stability and natural frequency. In MatLab:

```
% Compute eigenvalues of A
eigenvalues_A = eig(A);

% Display eigenvalues
disp('Eigenvalues of the state matrix A:');
disp(eigenvalues_A);
```

Figure 2: Computing Eigenvalues of A

- Stability

- If all eigenvalues have negative real parts, the system is stable (all modes decay over time).
- If any eigenvalue has a positive real part, the system is unstable (it diverges).
- If an eigenvalue has zero real part, the system has marginal stability (e.g., oscillations that don't decay).

- Frequency and Damping

- Complex eigenvalues ($\lambda = \sigma \pm iw$) indicate oscillatory modes, where σ(real part) represents the damping factor (how fast the oscillation decays) and w(imaginary part represents the natural frequency of oscillation.
- Real eigenvalues indicate non-oscillatory modes (purely exponential decay or growth).

In our case eigenvalues are:

$$\lambda_{1,2} = -2.8157 \pm 2.0067i$$

$$\lambda_{3,4} = -0.0073 \pm 0.0659i$$

All eigenvalues have negative real parts, which means the system is stable (all modes will decay over time).

Aircraft longitudinal motion consists of two primary modes:

- Short-period mode (fast, rapid pitch oscillation).
- Long-period mode (slow speed-altitude oscillation).

Short-Period Mode (Fast Oscillation)

The eigenvalues corresponding to this mode are λ_1 and λ_2 and the associated natural frequency is:

$$w_{n,SP} = \sqrt{(-2.8157)^2 + (2.0067)^2} = 3.4563\, rad/s$$

while the damping ratio is:

$$\zeta_{SP} = \frac{|\sigma|}{w_n} = \frac{2.8157}{3.4563} = 0.8145$$

So, this mode has a high frequency (fast oscillation) and it is well damped, meaning the oscillation decay quickly. Therefore, it corresponds to fast pitch oscillations and is typically associated with pilot-induced maneuvers (e.g., quick corrections in pitch attitude). It stabilizes within a few seconds.

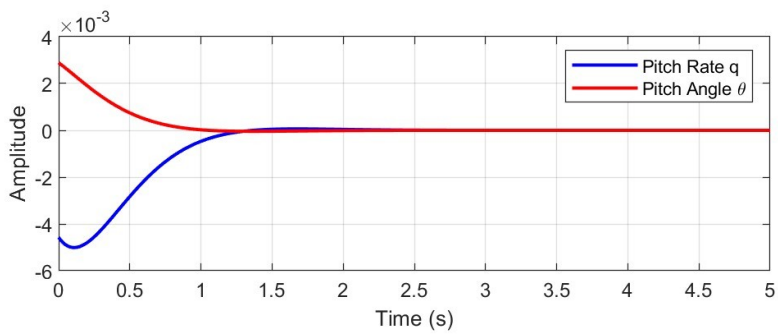

Figure 3: Short-Period Pitch Oscillations

Long-Period Mode (Slow Oscillation)

The eigenvalues corresponding to this mode are λ_3 and λ_4 and the associated natural frequency is:

$$w_{n,LP} = \sqrt{(-0.0073)^2 + (0.0659)^2} = 0.0663\, rad/s$$

while the damping ratio is:

$$\zeta_{LP} = \frac{|\sigma|}{w_n} = \frac{0.0073}{0.0663} = 0.1101$$

This mode has low frequency and it's very lightly damped meaning it takes a long time to decay. So, corresponds to slow oscillations in velocity and altitude. It is associated with energy exchange between kinetic and potential energy, like a gentle climb and descent. The damping is weak, meaning the oscillation persists for a long time before stabilizing.

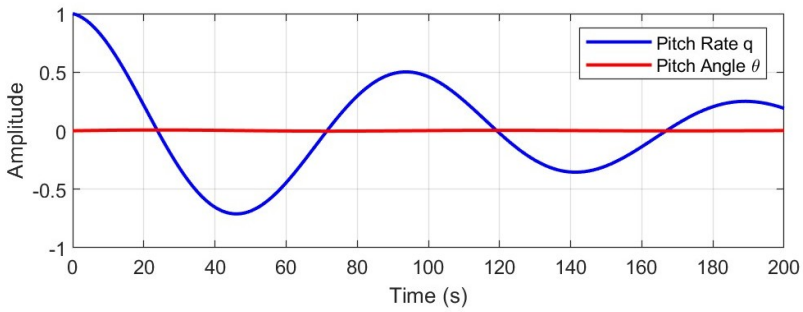

Figure 4: Long-Period Oscillations

2 Aircraft Simulation in Simulink

We'll now set up a Simulink model to simulate the linear aircraft system using a State-Space block. The goal is to observe the time evolution of the states v, w, q, θ while keeping the elevator angle δ_e constant.

As shown in Fig.5, since the elevator input δ_e is constant, we have no significant change in pitch rate q or pitch angle θ. Velocity v and vertical speed w remain steady with short transients initially. Hence, the system remains at trim conditions (as expected from the linearization around a steady flight mode).

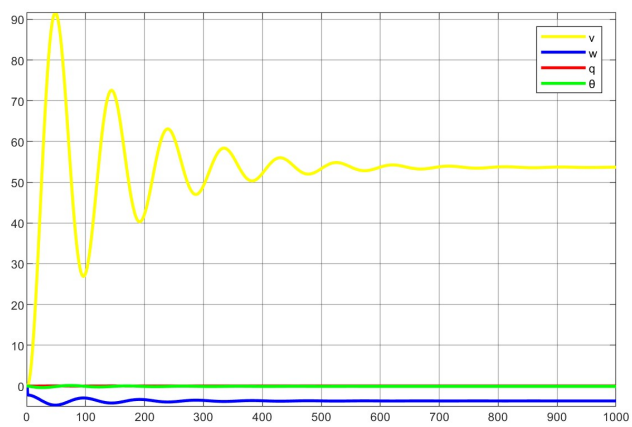

Figure 5: State-Space variables with constant elevator angle

Now, we will modify the input to be a sine wave representing the elevator angle δ_e with an amplitude of $0.1\,rad$ and a frequency range of $0.022\,rad/s$ to $0.099\,rad/s$. We will then analyze how pitch angle θ and longitudinal speed v respond to different input frequencies.

Response at Low Frequency ($f = 0.022\, rad/s$)

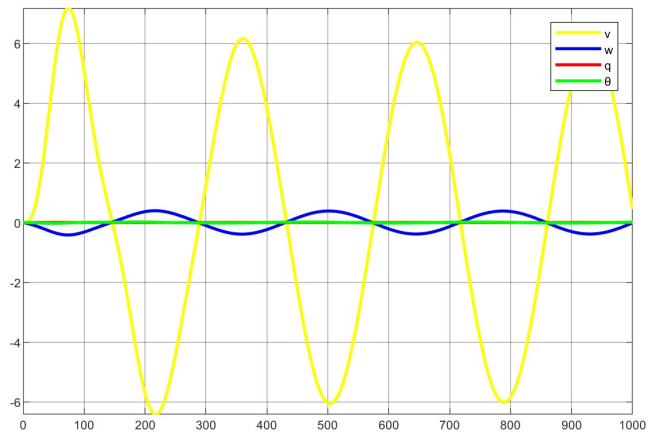

Figure 6: State-Space variables with low-frequency sinusoidal elevator angle

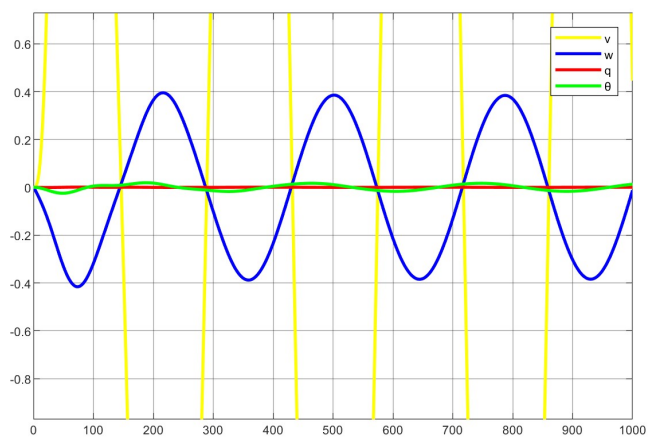

Figure 7: Detail of Figure 6

Pitch angle θ follows the input closely, with a nearly in-phase response. Longitudinal speed v also oscillates, but at smaller amplitude. This means that the aircraft follows the low-frequency elevator deflection smoothly.

Response at Moderate Frequency ($f = 0.05\,rad/s$)

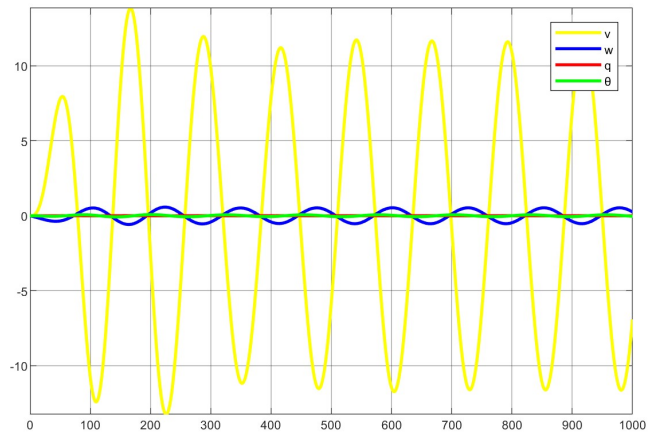

Figure 8: State-Space variables with moderate frequency sinusoidal elevator angle

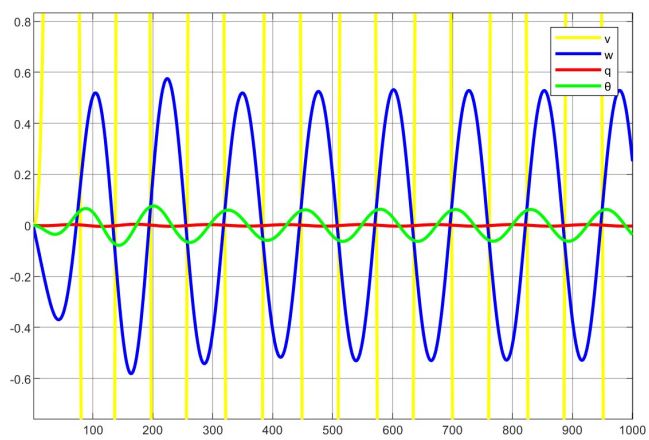

Figure 9: Detail of Figure 8

Pitch angle θ in this case increased its oscillation amplitude but it shouldn't be the case when increasing frequency of the input; this happens because when the frequency f is close to the natural frequency of one of the natural modes (in this case: $w_{n,LP} = 0.0663\,rad/s$), a resonance-like effect can occur, leading to increased amplitude of the response. Furthermore longitudinal speed v oscillates more noticeably and a phase lag starts appearing between θ and the input. So, the aircraft starts lagging the input more as frequency increases.

Response at High Frequency ($f = 0.099\, rad/s$)

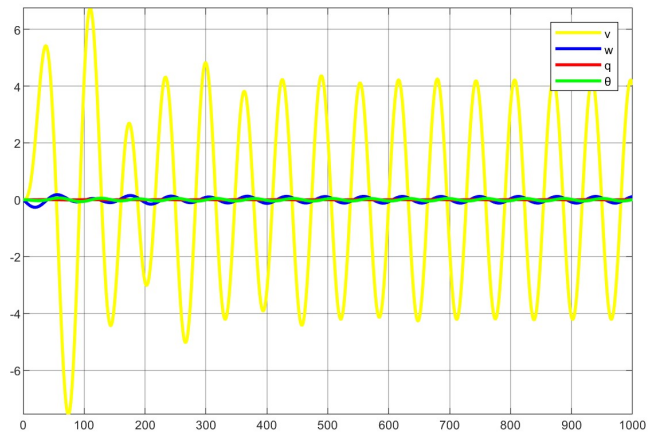

Figure 10: State-Space variables with high-frequency sinusoidal elevator angle

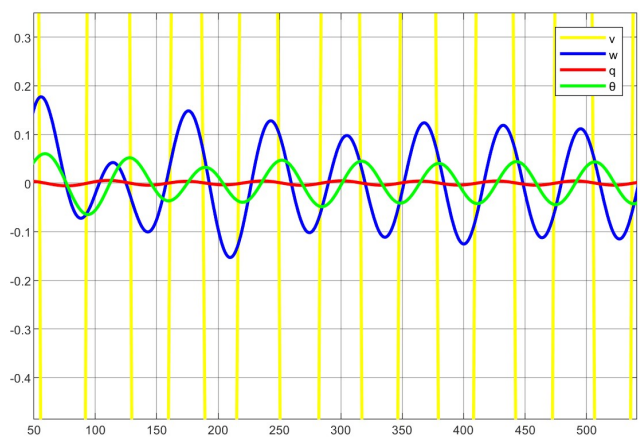

Figure 11: Detail of Figure 10

Pitch angle θ has a smaller amplitude. Longitudinal speed v barely responds and there's a significant phase lag between δ_e and θ. Therefore, at higher frequencies, the aircraft's short-period dynamics dominate. Less response in v and θ means the aircraft cannot follow fast control inputs effectively. This is due to aerodynamic inertia and control surface limitations.

3 Flight control - Pole placement control

We aim to design a state feedback controller $u = -KX$ to place the system poles in stable locations, preferably without oscillations. To eliminate oscillations, we choose real poles and to ensure stability, we choose them negatives either. Moreover, we choose them sufficiently far left in the complex plane to avoid excessive sensitivity. The new closed-loop system poles (eigenvalues of the "perturbed" matrix A) should match the chosen poles.

Let's first compute K manually and then check the result on MatLab, if it's correct, we can proceed to compute the new eigenvalues of A which now is given by:

$$A_{new} = A - BK$$

To compute K, we use the Ackermann's formula. So, let's first compute the controllability matrix

$$\mathcal{C} = [B \quad AB \quad A^2B \quad A^3B]$$

The system is controllable if $\mathcal{C}$ is full rank.

$$\mathcal{C} = \begin{bmatrix} 0 & 0.001365 & 1.1439 & -3.6562 \\ 0.1351 & -27.1534 & 151.2670 & -527.2962 \\ -0.1446 & 0.5266 & -1.2372 & 0.6729 \\ 0 & -0.1446 & 0.5266 & -1.2372 \end{bmatrix}$$

Since $\det(\mathcal{C}) \neq 0$, it has full rank and the system is controllable.

Ackermann's formula directly computes K from:

$$K = [0 \quad 0 \quad 0 \quad 1]\mathcal{C}^{-1}\Phi(A)$$

where $\Phi(A) = A^4 + a_1A^3 + a_2A^2 + a_3A + a_4$. To find a_1, a_2, a_3, a_4 we must solve:

$$(p - P_1)(p - P_2)(p - P_3)(p - P_4) = 0$$

where P_1, P_2, P_3, P_4 are the desired poles. So in our case we have:

$$(p + 1)(p + 2)(p + 3)(p + 4) = 0$$

$$\Longrightarrow p^4 + 10p^3 + 35p^2 + 50p + 24 = 0$$

and finally:

$$\Phi(A) = A^4 + 10A^3 + 35A^2 + 50A + 24$$

$$\Longrightarrow K = [0 \quad 0 \quad 0 \quad 1]\mathcal{C}^{-1}\Phi(A) = [8.2813 \quad 0.2195 \quad -29.9146 \quad -138.8872]$$

Let's now verify this result on MatLab and let's also check if the poles are placed correctly:

```
%   Stable poles
poles = [-1, -2, -3, -4];

% Compute feedback gain K using pole placement
K = place(A, B, poles);

% Display the result
disp('State feedback gain matrix K:');
disp(K);


% Compute closed-loop system matrix
A_new = A - B * K;

% Create new state-space system
sys_cl = ss(A_new, B, C, D);

% Display new eigenvalues (should match desired_poles)
eig_new = eig(A_new);
disp('Eigenvalues of closed-loop system:');
disp(eig_new);
```

```
State feedback gain matrix K:
    8.2813    0.2195  -29.9146 -138.8872

Eigenvalues of closed-loop system:
   -4.0000
   -1.0000
   -3.0000
   -2.0000
```

Figure 12: Poles placement and stability check

Since the eigenvalues of closed-loop system correspond to the chosen poles, K will modify the system dynamics so that the closed-loop system has poles at $[-1, -2, -3, -4]$.

Now, we want to add a step function (representing a perturbation u_p to the state feedback control law ($u = u_p - KX$). This helps analyze how the aircraft responds to external disturbances and how well the flight control system compensates for them. We set for the simulation a stop time of $10s$ so that we can clearly see the transient effects (Fig.13).

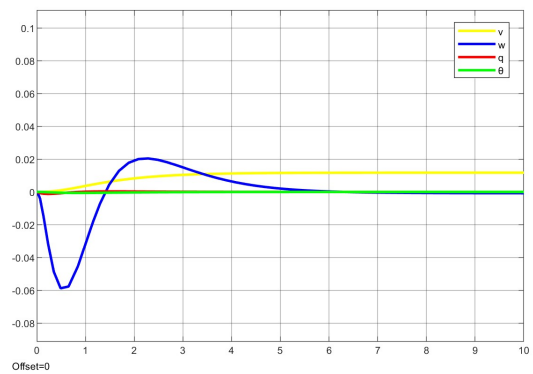

Figure 13: System with Perturbation Input and poles at -1,-2,-3,-4

After the step input is applied, the pitch angle θ and velocity v will deviate due to the external perturbation. The controller will react to bring them back to equilibrium. Without control, the aircraft would deviate significantly due to perturbation while with feedback control $(-KX)$, the system corrects the deviation, stabilizing the aircraft.

If the poles are further to the left we can see less transient effects (Fig.14) and this usually means that the system reaches its equilibrium quickly and with little oscillation , which is generally a good thing. However, if the system is reaching equilibrium too quickly and requires large control inputs, one might wants to balance the control effort and response speed.

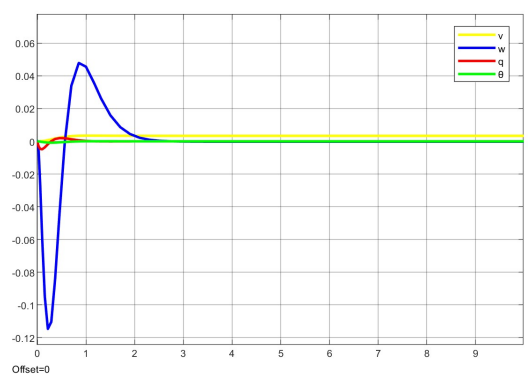

Figure 14: System with Perturbation Input and poles at -4,-5,-6,-7

Even though the system stabilizes faster with poles placed further left, it's important to make sure that the magnitude of the control input stays within reasonable limits. If the control input is too large, it might not be feasible in a real-world system, where there are physical limits on control deflections (like the elevator angle).